Varunprasaath R. Selvaraju, Ph.D.

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PROFESSIONAL SUMMARY

PhD chemist/materials scientist with 6+ years developing and characterizing functional materials across polymer, thin-film, and electrochemical systems, using DOE, spectroscopy, thermal analysis, and structure-property analysis to move lab-scale experiments toward reproducible process windows. Currently leading bio-based polymer process-development work and bringing hands-on synthesis, failure-mode analysis, and lab operations experience relevant to advanced materials scale-up.

CORE COMPETENCIES

Materials Synthesis & Development | Structure-Property-Processing | DOE/SPC | Spectroscopy & Thermal Analysis | Electrochemical Materials | Process Scale-Up | Analytical Method Execution | Lab Safety & SOPs

WORK EXPERIENCE

University of Colorado Boulder
July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided process-development experiments for bio-based polymer systems, defining process windows, material-quality readouts, and reproducible specifications for scale-up.
- Execute hands-on synthesis and characterization workflows using spectroscopy, FT-IR, and thermal analysis to connect processing conditions with material structure and performance.
- Translate experimental datasets into SOP-ready recommendations, technical reports, and safety-conscious laboratory workflows for repeatable execution.

University of Colorado Boulder
July 2021 - July 2025

Graduate Researcher

- Designed, synthesized, and evaluated 15+ organic and organometallic electrochemical material variants, building structure-property datasets from spectroscopic, electrochemical, and stability measurements.
- Identified material failure modes including oxidative degradation, interface delamination, and redox instability, then used root-cause analysis to recommend design and process mitigations.
- Applied multi-technique characterization including UV-Vis-NIR, FT-IR, TGA, DSC, cyclic voltammetry, spectroelectrochemistry, NMR, and GPC to evaluate material performance and reproducibility.

Georgia Institute of Technology
Aug 2019 - July 2021

Graduate Researcher

- Co-led AFOSR-funded process development for molecular thin-film coatings, developing surface-preparation protocols and CVD-based deposition strategies to control film morphology and device performance.
- Connected coating process conditions with optical/material performance using spectroscopy, thin-film characterization, and iterative experimental design.

ETH Zurich
May 2017 - July 2017

Research Assistant

- Developed and optimized polymer thin-film deposition processes using DOE strategies and in-situ QCM-D process monitoring to characterize film-growth kinetics and improve coating uniformity.

University of Colorado Boulder
2021 - 2025

Laboratory Operations Manager

- Managed operations for a 15-member process-development laboratory, implementing equipment tracking, calibration routines, SOPs, and safety protocols that improved instrument uptime by 30%.
- Trained researchers on instrument workflows, experimental documentation, and safe handling practices while balancing multiple concurrent materials-development projects.

Georgia Tech
2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Advanced Materials Synthesis & Process Scale-Up — Materials Development

Designed synthesis/process experiments for functional molecular and polymer materials, using DOE and multi-technique characterization to define reproducible process windows and performance readouts.

DOE, SPC, FT-IR, UV-Vis-NIR, TGA, DSC, NMR, GPC, electrochemistry

Structure-Property Characterization of Electrochemical Materials — Characterization

Built datasets across 15+ electrochromic material variants to connect molecular design, redox behavior, optical response, and degradation mechanisms.

Cyclic voltammetry, spectroelectrochemistry, UV-Vis-NIR, FT-IR, thermal analysis

AFOSR Defense Thin-Film Coatings — Thin Films

Developed CVD/surface-preparation workflows for molecular coatings, translating process controls into film morphology and device-performance outcomes.

CVD, surface preparation, spectroscopy, thin-film process development, QCM-D

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder 2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials
Advisor: Prof. Seth Marder
(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc) 2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Materials Development & Synthesis: Organic and organometallic functional materials, polymer systems, thin films, coatings, surface functionalization, CVD, process optimization, lab-to-scale-up planning

Analytical & Characterization: FT-IR, UV-Vis-NIR, ellipsometry, TGA, DSC, cyclic voltammetry, spectroelectrochemistry, QCM-D, NMR, GPC, GC-MS, LC-MS

Process & Data: DOE, SPC, JMP (basic), Python (basic), Excel, statistical data analysis, process windows, reproducibility studies, technical reports

Reliability & Manufacturing Readiness: Failure mode identification, root cause analysis, degradation/stability testing, SOPs, calibration, safety protocols, technical documentation

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." Chemistry of Materials, 2025, 37, 5558-5568.
2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." RSC Advances, 2018, 8, 20048-20055.
3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.

Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.

Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.

Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)